Geometric perspectives on the Analysis of Learning Algorithms

Forewords on the scope of the proposal. Machine Learning (ML) stands as a dynamic field at the intersection of applied mathematics and computer science, experiencing a recent explosion in growth. Understandably, considerable efforts are underway to harness this new tool for addressing long-standing problems in image processing, natural language or applied science. Yet, in the swift evolution of science, there is a simultaneous need to decelerate, fostering a comprehensive understanding and mastery of ML to establish it as a reliable and positive force. Notably, certain trends, such as training billion-parameter models with limited *alignment control*, appear contrary to the responsibility that science bears for ecological considerations and controllability. This mathematical proposal embodies the intent to comprehend and elevate ML into the realm of reliable science. This is the reason why, while being mainly theoretical, we believe that solving the questions raised by the proposal are instrumental if we, as a community, want to push further the (incredible) practical development of the field.

1. Context, positioning and objectives.

1.1. Position of the problem

Stochastic optimization in non-convex landscapes. Machine Learning has garnered significant attention in the past two decades, attracting both industry interest for data-driven decision problems and widespread recognition within the scientific community. This recent attention is certainly due to its ability to efficiently solve a wide class of high-dimensional problems with rapid and easy-to-implement algorithms. The core of its algorithmic success relies on the minimization of a high-dimensional loss function. Concretely, one has a model h_θ that depends on a high-dimensional parameter $\theta \in \mathbb{R}^p$ and a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$. The goal is to find a parameter θ that minimizes the loss

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell(h_\theta(x_i), y_i),$$

where ℓ is a loss function penalizing the difference between the prediction $h_\theta(x_i)$ (e.g. a neural network) and the true label y_i. Quite surprisingly, one of most widespread procedure to minimize this loss is simply the *Stochastic Gradient Descent* (SGD) [1]. This algorithm is based on a fairly simple update rule: at step $t \in \mathbb{N}$, choose at random a sample (x_{i_t}, y_{i_t}) from the dataset and update the parameter θ as $\theta_{t+1} = \theta_t - \eta \nabla_\theta \ell(h_{\theta_t}(x_{i_t}), y_{i_t})$, where $\eta > 0$ is the step size. This is the building block of any stochastic algorithm that underlies the training of machine learning models. Yet, despite its simplicity, its efficiency and versatility are not yet understood. Indeed, while in the case where $\mathcal{L}$ is convex, SGD is known to reach satisfactory solutions [2], the case of non-convex losses is still largely open. Obviously, the latter case is the one of interest in practice, as any parametrization with neural networks h_θ naturally leads to a non-convex loss $\mathcal{L}$.

General description of the objectives of the proposal. We are interested in the training of neural networks with SGD. Naturally, the first question that arises is whether SGD converges toward a global minimum of the loss $\mathcal{L}$, and if so, how fast? This question is treated as **Task I** in the proposal. Going further, one can ask whether the solution found by SGD is "good" in some sense, and if so, how is it possible to quantify this? These questions, arising from the overparametrization of current architectures, are at the core of the field of non-convex optimization and have been the subject of intense research in the past decade. This research track is still largely open, and the second aim of this proposal is to contribute to this understanding. More precisely, two factors can largely influence the behavior of the algorithm and the quality of the solution found: the *architecture* of the neural network h_θ (**Task II-a**) and the *noise* inherent of the SGD algorithm (**Task II-b**). We detail these two questions in the following section.

Framework. Obviously, addressing these questions in a general sense is a long-term goal. The concrete aim of this proposal is to first build mathematical frameworks in which each question can be understood: the convergence for simple neural networks will be addressed by deriving *quantitative* rates of convergence linked to the Wasserstein geometry induced by the (stochastic) gradient descent [3]. We will try to show that the second and third questions are highly dependent on the model through the vector field $(\nabla_\theta h_\theta)_\theta$, which characterizes the loss landscape the dynamics travels in. Geometry arises as the hidden key to understand the dynamics of the algorithm.

1.2. Objectives and methodology

Task I: Convergence of neural networks training. As recalled before, in the case of neural networks, due to the non-convexity of the objective, the convergence of the (stochastic) gradient descent is not solved. Even in the "simple case" of one hidden layer, the problem is still open, even if progress has been made in some settings. Indeed, the convergence has been addressed in two specific cases: the first one corresponds to an initialization of the algorithm that is arbitrarily large and has been named either the lazy regime [4] or the Neural Tangent Kernel regime [5]. However, it has been shown that this corresponds to a degenerate case where the neurons barely move [4] and the reached solution generalizes badly. The second case where convergence is proven, also known as *mean-field regime*, corresponds to an *infinite number of neurons* [6], for which the Wassertein geometry has a prominent role. Yet, neither any quantitative rate of convergence nor a quantitative theory for a finite number of neurons have been derived in this case.

We attempt to pursue this study, providing convergence rates for a large but finite number of neurons. In order to achieve this, we will need to track more precisely the dynamics of the algorithm, and in particular analyze thoroughly the geometry of the loss landscape. More precisely, we believe that this question can be decomposed in two steps: (i) obtain a quantitative convergence rate for the infinite-neurons limit: this can be done for example by showing that the loss landscape satisfies a Polyak-Łojasiewicz condition [7] along the trajectory; (ii) show a quantitative propagation of chaos result [8] for the finite number of neurons case, i.e., show that it does not deviate too much from the *mean-field* case. We will first address cases for which the geometry of the loss landscape is simple, for example in high-dimension and almost orthogonal inputs [9], and then extend these results to more complex and general cases. This work will be carried out in collaboration with one of the pioneers of this line of research: Lenaïc Chizat (EPFL).

Task II-a: Implicit bias: the role of the architecture. While the algorithms used for training have not changed much these last years (regularized SGD + Adam variants [10]), neural networks *architectures* have been at the core of recent improvements. As a simplified representation, we can envision this through a (non-linear!) timeline of models progressing in both complexity and performance

Kernel methods $\to$ Multi-Layer Perceptron $\to$ Convolutional Neural Networks$\to$ ResNets $\to$ Transformers.

Mathematically, what stands out is that all these architectures encode models of considerable expressiveness in a *similar* fashion. Despite intensive efforts to distinguish these models using approximation theory tools, this framework could not explain their success. On the contrary, one can see the difference between these models dynamically; as SGD updates are guided by the gradient of the function $\theta \mapsto h_\theta(x)$, to each parametrized family (i.e. each architecture) corresponds a *geometry* induced by the gradient field $(\nabla_\theta h_\theta(x))_\theta$. Understanding this field for state-of-the-art architectures like transformers is out of reach for now, but for simple models, this has been a fruitful line of research initiated by Nati Srebro [11]. In this way, every parametrized family is "associated" to an intrinsic regularizer, often referred to as capacity: diagonal neural network are associated to the ℓ_1-norm, matrix factorization to the nuclear-norm [12] and two layer (or more) neural-networks to variational norms [13]. Yet, once again, beyond the diagonal linear network case, this "association" is far from being precisely understood mathematically, hence the question: *What is the mathematical relationship between architecture and capacity control?*

In this research track, we will explore that a possible answer come from the field of *Information geometry* [14], that tries to characterize the natural geometry induced by a family of predictors. For diagonal networks, it is possible to map the system to a *dually flat geometry* equipped with a Kullback-Leibler divergence, but this re-mapping does not easily extend to the other settings of importance described above.

In fact, the existence of a Bregman divergence that underlies the dynamics of the algorithm is a key open question in the quest of capacity control. We aim at bridging this gap leveraging a *canonical way to define divergences on statistical manifolds* [15], and show that these so-constructed divergences may be the capacity control that explains the good solutions that are found by SGD.

In order to do this, we will try to characterize the implicit bias of models of increasing complexities. The case of diagonal linear networks has been already solved [16], but the first new step would be to handle the *matrix factorization case*. Finally we will try to extend these results to the case of two-layer neural networks, where the geometry is more complex. This work will be carried out with Joan Bruna (NYU).

Task II-b: The role of noise in SGD learning. Lately, it has been shown that in fact, the inherent noise of SGD is actually instrumental for the good performance of neural networks training [17]. This is a surprising result, as one would expect that the noise would only slow down the convergence of the algorithm. This remarkable feature has been investigated in a general context in [18] where it was shown that the presence of noise *locally* regularizes the loss landscape by some local norm. Lately, in a specific noise regime, in [19] Li et al. showed that the noise drives the dynamics toward very specific solutions in the late times of the dynamics. Yet, these are only local results in specific settings, and the general understanding of the role of noise in the training of neural networks is still a mystery.

In a series of works [20, 21, 22], we have developed a framework, relying on stochastic differential equation (SDE) models [23] to analyze the role of the noise in SGD learning. Indeed, we have shown that the parameters of the network typically evolve according the the effective SDE: $\mathrm{d}\theta_t = -\nabla_\theta \mathcal{L}(\theta_t)\mathrm{d}t + \sqrt{\eta}\phi(X,\theta_t)^\top \mathrm{d}B_t$, where $(B_t)_{t\geq 0}$ is a standard Brownian motion, $\phi(X,\theta) := [\nabla_\theta h_\theta(x_i)^\top]_{i=1}^n$ are the *NTK features* of the model and finally $\eta > 0$ is the step size. This representation of the dynamics has a concrete interpretation: the model is balanced between two "forces": one comes from the usual gradient through $-\nabla\mathcal{L}$, but the other force is inherently stochastic and corresponds to $\phi(X,\theta_t)^\top \mathrm{d}B_t$. For simple models like *diagonal linear networks* [16], we have shown that the noise term is responsible for a sparse inducing implicit bias as the combination of the noise and the gradient result in a *Lasso program* [21].

Similarly to **Task II-a**, there is a natural series of models of increasing difficulties that can be tackled to test our framework. The first new step would be to show that the noise term enforces low rank solutions in the case of matrix factorization. This is a natural extension of the diagonal linear networks case, and we believe that the same tools can be used to show this. The second step would be to extend this simplicity bias in the case of two-layer neural networks. Obviously, due to the non-linearity of the predictor h_θ, this is a much more difficult problem, that will be carried out with the help of Julien Reygner (CERMICS), expert of longtime behavior of stochastic processes.

In the long term, and in order to develop a systematic way to solve this problem, we may resort to the *Quasi potential theory* from large deviation principles [24] when η is a small parameter. In fact in this limit of small η, quasi potential theory is a powerful framework that shows that the "regularized" loss we are looking for solves a Hamilton-Jacobi equation that results from the combination of the gradient and noise parts.

2. Partnerships and collaborations

The coordinator of this project is Loucas Pillaud-Vivien, whose work focuses on the theoretical analysis of machine learning optimization algorithms. He defended his PhD thesis at Inria/Ens Paris in October 2020, after which he was a postdoc at EPFL (2020-2022) and a Courant Instructor at NYU (2022-2023). Since January 2024, he has been a permanent researcher at Cermics, École des Ponts. He will devote 70% of his time to this project.

Optimization is at the heart of the three research tracks of the proposal but each one carries its own specificity. The first track requires some knowledge in neural networks training, which has already been investigated by Loucas Pillaud-Vivien [9], and its link with Wasserstein gradient flow, an approach that makes Lénaïc Chizat a perfect collaborator. The second track requires some knowledge in information geometry, mirror flow and calculus of variations, domain in which both Loucas Pillaud-Vivien and Joan Bruna already collaborated [25]. For the third track, the longtime behavior of stochastic processes are at the

heart of the problem, and this is the reason why we will tackle the problem with an expert of the domain: Julien Reygner [26].

The JCJC funding would allow to create a team working on these projects, by hiring a PhD student (Track I) and a postdoc (Track II-a), and leveraging collaborations with the international researchers mentioned above. During the period of the project, the funding would allow to organize a workshop in a French institution (CIRM or IHP) and gather the community working on the topics of the proposal.

Total Funding: 369 487,90€. Detail: 135.000€ for one PhD student, 117.840€ pour 2 years of Post-doc, 5.100€ for a 6 months master 2 internship, 6.600€ for the material, 3.000€ for publication fees, 45.000 € for missions, 13.000€ to organize a conference, 43.947,90€ for environmental fees.

References

[1] Léon Bottou, Frank E Curtis, and Jorge Nocedal. Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311, 2018.

[2] Léon Bottou. Stochastic gradient descent tricks. In *Neural Networks: Tricks of the Trade: Second Edition*, pages 421–436. Springer, 2012.

[3] Lenaic Chizat and Francis Bach. On the global convergence of gradient descent for over-parameterized models using optimal transport. *Advances in neural information processing systems*, 31, 2018.

[4] Lenaic Chizat, Edouard Oyallon, and Francis Bach. On lazy training in differentiable programming. *Advances in neural information processing systems*, 32, 2019.

[5] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. *NeurIPS*, 2018.

[6] Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018.

[7] Chaoyue Liu, Libin Zhu, and Mikhail Belkin. Loss landscapes and optimization in over-parameterized non-linear systems and neural networks. *Applied and Computational Harmonic Analysis*, 59:85–116, 2022.

[8] Alain-Sol Sznitman. Topics in propagation of chaos. *Ecole d'été de probabilités de Saint-Flour XIX—1989*, 1464:165–251, 1991.

[9] Etienne Boursier, Loucas Pillaud-Vivien, and Nicolas Flammarion. Gradient flow dynamics of shallow relu networks for square loss and orthogonal inputs. *NeurIPS*, 35:20105–20118, 2022.

[10] Diederik P. Kingma. Adam: A method for stochastic optimization. *Internation Conference of Learning and Representation*, 2014.

[11] Behnam Neyshabur, Ryota Tomioka, and Nathan Srebro. In search of the real inductive bias: On the role of implicit regularization in deep learning. *International Conference on Learning Representations*, 2014.

[12] Suriya Gunasekar, Blake E Woodworth, Srinadh Bhojanapalli, Behnam Neyshabur, and Nati Srebro. Implicit regularization in matrix factorization. *NeurIPS*, 30, 2017.

[13] Francis Bach. Breaking the curse of dimensionality with convex neural networks. *JMLR*, 2017.

[14] Shunichi Amari. *Information geometry and its applications*, volume 194. Springer, 2016.

[15] Nihat Ay and Shun-ichi Amari. A novel approach to canonical divergences within information geometry. *Entropy*, 2015.

[16] Blake Woodworth, Suriya Gunasekar, Jason D Lee, Pedro Savarese, Daniel Soudry, and Nathan Srebro. Kernel and rich regimes in overparametrized models. In *COLT*, 2020.

[17] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 2021.

[18] Alex Damian, Tengyu Ma, and Jason D Lee. Label noise sgd provably prefers flat global minimizers. *Advances in Neural Information Processing Systems*, 34:27449–27461, 2021.

[19] Zhiyuan Li, Tianhao Wang, and Sanjeev Arora. What happens after sgd reaches zero loss?–a mathematical framework. *International Conference on Learning Representations*, 2021.

[20] Scott Pesme, Loucas Pillaud-Vivien, and Nicolas Flammarion. Implicit bias of sgd for diagonal linear networks: a provable benefit of stochasticity. *NeurIPS*, 34:29218–29230, 2021.

[21] Pillaud-Vivien Loucas, Julien Reygner, and Nicolas Flammarion. Label noise (stochastic) gradient descent implicitly solves the lasso for quadratic parametrisation. In *COLT*, 2022.

[22] Maksym Andriushchenko, Aditya Vardhan Varre, Loucas Pillaud-Vivien, and Nicolas Flammarion. Sgd with large step sizes learns sparse features. In *ICML*, 2023.

[23] Qianxiao Li and Cheng Tai. Stochastic modified equations and dynamics of stochastic gradient algorithms I: Mathematical foundations. *J. Mach. Learn. Res.*, 20:40–1, 2019.

[24] MI Freidlin and AD Wentzell. *Random perturbations*. Springer, 1998.

[25] Alberto Bietti, Joan Bruna, and Loucas Pillaud-Vivien. On learning gaussian multi-index models with gradient flow. *arXiv preprint arXiv:2310.19793*, 2023.

[26] Freddy Bouchet and Julien Reygner. Generalisation of the eyring–kramers transition rate formula to irreversible diffusion processes. In *Annales Henri Poincaré*, volume 17, pages 3499–3532. Springer, 2016.